

Part I Report and Work Order: Fixed-State ADAPT-VQE Gradient-Measurement Diagnostics

Corrected method definitions and preliminary oracle results

September 2026

Abstract

This report defines and compares three measurement strategies for the generator-selection step in gradient-based ADAPT-VQE. The fixed method names are **M1: FC-UG**, fully commuting universal groups for static all-gradient estimation; **M2: BAI-FC-IG**, best-arm identification using fully commuting individual-gradient measurements; and **M3: BAI-FC-UG**, best-arm identification using fixed parent fully commuting universal groups. These names must be used consistently in text, code, tables, plots, and file names. The reported numerical values are oracle/noiseless planning diagnostics: exact gradients and exact fragment variances are used to estimate measurement requirements. They are not empirical finite-shot simulations. Part III symmetry filtering and tapering are not included in the Part I numerical tables; Part III supplies a smaller input problem to the same measurement methods.

Revision of September 2026. All numbers in this revision are produced by a single from-geometry implementation in which each method is a separate module that saves the fully commuting groups and the fragment variances behind every table entry. Every run passes a validation gate requiring the Jordan–Wigner Hamiltonian to reproduce the PySCF RHF energy on the Hartree–Fock determinant and the PySCF FCI energy as its lowest eigenvalue in the target particle-number sector. All nine cases are now complete, H₂O included, so the three-method comparison no longer has missing entries. Two conclusions of the previous revision do not survive and are corrected in Sec. 16. Both free parameters of the previous revision have been removed: M1 is now evaluated at the target radius that matches the winner-identification task, and the BAI methods are evaluated in the schedule-free exact limit rather than on a geometric radius schedule. This revision adds two further pieces. A no-sharing baseline completes the two-by-two the methods span and makes the contribution of each mechanism separately attributable (Sec. 10). A finite-shot study replaces the exact gradients with estimates from a finite number of shots and runs the same elimination rule on those (Sec. 11); it retracts one conclusion of the earlier revision and shows that the confidence convention is far stricter than the stated failure probability requires.

1 One-sentence problem

Given a molecular Hamiltonian H , a current ADAPT-VQE state $|\psi\rangle$, and a UCCSD generator pool $\{G_i\}$, estimate the measurement cost required to identify a generator with the largest absolute gradient

$$i^* = \arg \max_i |g_i|, \quad g_i = \langle \psi | [H, G_i] | \psi \rangle. \quad (1)$$

The scientific question is whether using winner-identification statistics and shared fully commuting measurement contexts reduces the gradient-selection measurement cost relative to static all-gradient estimation.

2 Fixed method names

The phrase “naive/no-sharing” is retired. It referred only to a primitive diagnostic and is not one of the project methods. Use only the method names in Table 1.

Table 1: Fixed method names.

Name	Full name	Definition
M1: FC-UG	Fully Commuting Universal Groups	Build one global parent FC grouping for the union of all Pauli products appearing in all ADAPT gradient commutators. Estimate the full gradient vector.
M2: BAI-FC-IG	BAI with Fully Commuting Individual Gradients	Treat each generator gradient as a BAI arm. For each active gradient, use an FC grouping of that individual commutator. No cross-gradient universal sharing is used.
M3: BAI-FC-UG	BAI with Fully Commuting Universal Groups	Use the same fixed parent FC universal groups as M1, but apply BAI elimination and sample only parent contexts needed to distinguish surviving generators.

3 Object lineage

The qubit Hamiltonian is

$$H = \sum_p h_p P_p, \quad (2)$$

where P_p are Pauli products after Jordan–Wigner mapping. For each generator G_i , define the gradient operator

$$C_i = [H, G_i]. \quad (3)$$

The ADAPT gradient is the expectation value of this operator,

$$g_i = \langle C_i \rangle_\psi. \quad (4)$$

Expanding all gradient operators in one shared Pauli-coordinate system gives

$$C_i = \sum_\ell A_{i\ell} R_\ell, \quad \mu_\ell = \langle R_\ell \rangle_\psi, \quad \mathbf{g} = A\boldsymbol{\mu}. \quad (5)$$

The parent universal Pauli support at this ADAPT step is

$$\mathcal{B}_0 = \bigcup_i \text{supp}(C_i). \quad (6)$$

M1 and M3 use $\mathcal{B}_0$ as a shared measurement object. M2 does not; it groups each C_i separately.

4 The three algorithms

4.1 M1: FC-UG

M1 estimates the full gradient vector. It forms one global FC grouping of the parent support,

$$\mathcal{B}_0 \longrightarrow \mathcal{M}_1, \dots, \mathcal{M}_K, \quad (7)$$

where each $\mathcal{M}_\alpha$ is a fully commuting measurement context. Measurements of these contexts estimate the shared Pauli expectations μ_ℓ and reconstruct every gradient using Eq. (5).

The pilot implementation used deterministic greedy FC grouping of $\mathcal{B}_0$. This is a Barnes-style global FC all-gradient baseline because it shares contexts across the whole ADAPT gradient pool, but it is not claimed to reproduce every specialized construction in Refs. [3, 2].

4.2 M2: BAI-FC-IG

M2 changes the statistical target from full-vector estimation to winner identification. Each candidate gradient is an arm in a best-arm-identification problem. Its own commutator support is grouped separately,

$$\text{supp}(C_i) \longrightarrow \mathcal{M}_{i,1}, \dots, \mathcal{M}_{i,K_i}. \quad (8)$$

BAI maintains confidence intervals for active gradients and eliminates a generator i if

$$|\widehat{g}_i| + r_i < \max_{j \in \mathcal{A}_r} (|\widehat{g}_j| - r_j), \quad (9)$$

where $\mathcal{A}_r$ is the active generator set at BAI round r and r_i is the current confidence radius. M2 does not exploit universal cross-gradient sharing.

Implementation status. M2 is now implemented as its own module. Each run writes the individual-gradient FC grouping $\mathcal{M}_{i,k}$ for every generator, the oracle fragment standard deviation of every fragment, the per-arm shot cost, and the full BAI elimination history with confidence radii and active sets. The M2 column is therefore auditable; it remains an oracle planning diagnostic rather than a finite-shot result.

4.3 M3: BAI-FC-UG

M3 combines the mechanisms of M1 and M2. It starts with the same fixed parent universal FC contexts as M1,

$$\mathcal{M}_1, \dots, \mathcal{M}_K, \quad (10)$$

but BAI eliminates generator candidates,

$$\mathcal{A}_0 \supset \mathcal{A}_1 \supset \mathcal{A}_2 \supset \dots. \quad (11)$$

At BAI round r , the active Pauli support is

$$\mathcal{B}_r = \bigcup_{i \in \mathcal{A}_r} \text{supp}(C_i), \quad \mathcal{B}_r \subseteq \mathcal{B}_0. \quad (12)$$

The active measurement contexts are restricted parent contexts,

$$\mathcal{M}_\alpha^{(r)} = \mathcal{M}_\alpha \cap \mathcal{B}_r. \quad (13)$$

The parent contexts are not rebuilt during BAI. This nested construction keeps previous measurements compatible while allowing the algorithm to stop sampling information that only affects eliminated generators.

5 Oracle shot proxy

All Part I numbers are oracle/noiseless planning estimates. The state $|\psi\rangle$ is known exactly in the simulation, so the exact gradient values and exact fragment variances are available. No empirical finite-shot variance learning is included.

For one gradient operator written as a sum of independently measured FC fragments,

$$C_i = \sum_{\alpha=1}^{K_i} F_{i\alpha}, \quad (14)$$

with oracle fragment variances

$$\sigma_{i\alpha}^2 = \text{Var}_\psi(F_{i\alpha}) = \langle F_{i\alpha}^2 \rangle_\psi - \langle F_{i\alpha} \rangle_\psi^2, \quad (15)$$

the optimal independent-fragment allocation gives the approximate total number of shots needed to estimate g_i with one-standard-error target ϵ_i as

$$M_i(\epsilon_i) \approx \frac{\left(\sum_{\alpha=1}^{K_i} \sigma_{i\alpha}\right)^2}{\epsilon_i^2}. \quad (16)$$

When a confidence factor is used, ϵ_i is replaced by ϵ_i/z_{δ_i} . For BAI diagnostics the tolerance for an arm is determined by the current separation between its estimated absolute gradient and the best surviving absolute gradient. The pilot used a normal confidence model with family-wise failure probability $\delta = 0.05$.

For M2 this formula is applied separately inside each gradient arm, because M2 measures each arm in its own contexts. M1 and M3 instead share contexts between generators, so the allocation over contexts is not separable and is obtained by solving the convex programme

$$\min_{n_\alpha \geq 0} \sum_\alpha n_\alpha \quad \text{subject to} \quad \text{Var}(\hat{g}_i) = \sum_\alpha \frac{\sigma_{i\alpha}^2}{n_\alpha} \leq \epsilon^2 \quad \text{for every active } i, \quad (17)$$

where n_α is the number of shots taken in context α . Its stationarity condition is $n_\alpha = \sqrt{\sum_i \lambda_i \sigma_{i\alpha}^2}$ for multipliers $\lambda_i \geq 0$, which is solved by a damped multiplicative fixed-point iteration and then rescaled so that the tightest constraint holds exactly. The returned allocation is therefore feasible by construction, and for a single gradient it reduces exactly to Eq. (16). A previous envelope heuristic, which spread shots in proportion to $\max_i \sigma_{i\alpha}$, overestimated the M1 and M3 costs by up to a factor of eight and inverted the M2/M3 comparison; it has been replaced.

The BAI methods are evaluated in the schedule-free limit. As the common confidence radius r decreases continuously, generator i leaves the active set the moment $2r$ falls below its absolute-gradient deficit

$$\Delta_i = |g_1| - |g_i|, \quad (18)$$

so it is resolved to exactly $\Delta_i/2$; the leader survives to gap/2. Because measurements are reusable, an arm that leaves at radius r has paid the cost of being resolved to r and nothing more, and a shared context has paid the largest requirement it ever saw. The active set changes only at the finitely many distinct values of Δ_i , so the whole trajectory is determined by evaluating the allocation at those radii. For M2 this collapses to the closed form of Eq. (21). This replaces the geometric radius schedule of the previous revision, whose shrink factor was a free parameter and which, being coarse, overshoot the elimination points; Sec. 12 quantifies the difference.

Because these formulae depend on the exact grouping and fragment variances, every production run saves the FC groups, the fragment variances and the BAI histories used to generate each table entry.

6 Benchmark assumptions

The generator universe is the standard spin-orbital UCCSD pool conserving electron number N_e and S_z , but not generally total spin S^2 . No frozen-core approximation, no qubit tapering, and no Part II variance-aware coefficient splitting are included in the Part I numerical tables. The CISD state is used as a correlated proxy for a later ADAPT state; it is not a saved ADAPT trajectory snapshot.

FC group counts are routine-dependent. Pre-taper and post-taper counts, or any two rows in the same table, are comparable only if produced by the same deterministic FC grouping routine, with the same Pauli ordering, graph-coloring order, tie-breaking rule, and random seed. Every row in this revision

uses one routine: deterministic greedy first-fit grouping, Pauli strings sorted by decreasing weight and then lexicographically, a candidate joining the lowest-numbered group that accepts it, no random seed. The counts therefore differ from the previous revision, whose H_4 and H_2O rows were produced by a different partitioning routine; those rows were not comparable with the LiH row.

Validation gate. Every case is checked before any gradient is computed: the Jordan–Wigner Hamiltonian must reproduce the PySCF RHF energy on the Hartree–Fock determinant and the PySCF FCI energy as its lowest eigenvalue in the (N_α, N_β) sector, both to 10^{-8} Ha. Observed errors are at the 10^{-15} level. A case that fails the gate aborts rather than producing numbers.

Non-zero gradient threshold. Gradients that vanish by symmetry appear at the 10^{-8} numerical noise floor of the integral and SCF pipeline, while the smallest physically non-zero gradient in these cases is of order 10^{-3} . The counts below use a threshold of 10^{-6} , which lies on the plateau between the two. The threshold is not innocuous at tighter values: for LiH the count is 26 at 10^{-6} and 34 at 10^{-9} , and every run records the count at 10^{-4} , 10^{-6} , 10^{-8} and 10^{-9} so the choice stays visible.

7 Observable structure for all systems

Table 2: Global universal support used by M1: FC-UG and M3: BAI-FC-UG. The FC contexts are global parent contexts for $\mathcal{B}_0$, not per-gradient groupings, and all rows use the single greedy routine fixed in Sec. 5. Non-zero gradient counts use the 10^{-6} threshold.

Case	State	Qubits	Pool	Universal terms	FC-UG contexts	Nonzero g_i
LiH , $R=3.0$ Å	HF	12	92	30,196	980	26
H_4 square, side=1.0 Å	HF	8	26	1,276	53	10
H_4 square, side=1.0 Å	CISD	8	26	1,276	53	10
H_4 square, side=2.0 Å	HF	8	26	1,276	53	10
H_4 square, side=2.0 Å	CISD	8	26	1,276	53	10
H_2O eq., $R(\text{OH})=0.9572$ Å	HF	14	140	81,566	2,366	40
H_2O eq., $R(\text{OH})=0.9572$ Å	CISD	14	140	81,566	2,366	48
H_2O stretch, $R(\text{OH})=1.75$ Å	HF	14	140	81,566	2,356	40
H_2O stretch, $R(\text{OH})=1.75$ Å	CISD	14	140	81,566	2,356	48

The universal supports reproduce the earlier pilot data exactly (30,196 for LiH , 1,276 for H_4 , 81,566 for H_2O), as do the gradient values and gaps in Table 3 to six or more digits. The FC context counts differ because they now all come from one routine.

8 Gradient-selection difficulty

Table 3: Top-gradient gaps and M2: BAI-FC-IG oracle shot costs. Smaller gaps make winner identification harder; the BAI cost scales as the inverse square of the relevant deficit.

Case	State	Top generator	$ g_1 $	$ g_2 $	Gap	M2: BAI-FC-IG
LiH , $R=3.0$ Å	HF	D 2,3 $\rightarrow$ 10,11	0.243187	0.146799	0.096388	1,140,583
H_4 square, side=1.0 Å	HF	D 2,3 $\rightarrow$ 6,7	0.295316	0.273776	0.021539	854,629
H_4 square, side=1.0 Å	CISD	D 0,1 $\rightarrow$ 6,7	0.142467	0.024386	0.118081	102,806
H_4 square, side=2.0 Å	HF	D 2,3 $\rightarrow$ 6,7	0.315036	0.298387	0.016649	422,746
H_4 square, side=2.0 Å	CISD	D 0,1 $\rightarrow$ 6,7	0.098961	0.047102	0.051859	86,857
H_2O eq., $R(\text{OH})=0.9572$ Å	HF	D 4,5 $\rightarrow$ 12,13	0.304910	0.203820	0.101090	25,096,606
H_2O eq., $R(\text{OH})=0.9572$ Å	CISD	D 4,5 $\rightarrow$ 12,13	0.006925	0.004861	0.002064	46,457,815,707
H_2O stretch, $R(\text{OH})=1.75$ Å	HF	D 6,7 $\rightarrow$ 12,13	0.388493	0.366638	0.021856	17,423,631
H_2O stretch, $R(\text{OH})=1.75$ Å	CISD	D 6,7 $\rightarrow$ 12,13	0.106873	0.089917	0.016957	168,305,383

All three methods select the true maximum-gradient generator in all nine cases.

9 Complete three-method comparison

All three methods are compared at one criterion: the winner must be separated from the rest of the pool at family-wise confidence $\delta = 0.05$. M1 reaches that with a single common radius $\text{gap}/2$ over the whole pool; M2 and M3 reach it by elimination, resolving each generator only to $\Delta_i/2$. No free schedule or precision parameter remains in any column.

Table 4: Oracle context-shot comparison at the winner-identification criterion. $M1/M2 > 1$ means M2 is cheaper than M1; $M2/M3 > 1$ means M3 is cheaper than M2. Entries below unity are set in bold.

Case	State	M1: FC-UG	M2: BAI-FC-IG	M3: BAI-FC-UG	M1/M2	M2/M3	M1/M3
LiH, R=3.0 Å	HF	3,563,904	1,140,583	638,423	3.12	1.79	5.58
H4 square, side=1.0 Å	HF	2,244,862	854,629	566,951	2.63	1.51	3.96
H4 square, side=1.0 Å	CISD	59,642	102,806	53,727	0.58	1.91	1.11
H4 square, side=2.0 Å	HF	598,102	422,746	220,389	1.41	1.92	2.71
H4 square, side=2.0 Å	CISD	58,068	86,857	33,432	0.67	2.60	1.74
H2O eq., R(OH)=0.9572 Å	HF	172,147,028	25,096,606	22,130,852	6.86	1.13	7.78
H2O eq., R(OH)=0.9572 Å	CISD	411,524,732,152	46,457,815,707	44,971,732,317	8.86	1.03	9.15
H2O stretch, R(OH)=1.75 Å	HF	3,864,803,902	17,423,631	20,163,542	221.81	0.86	191.67
H2O stretch, R(OH)=1.75 Å	CISD	5,715,348,735	168,305,383	165,843,674	33.96	1.01	34.46

M3 is cheaper than M1 in every row, which it must be: every M3 active set is a subset of M1’s whole pool, measured in the same contexts at a radius no tighter than M1’s. M3 is cheaper than M2 in eight rows of nine. M2 is cheaper than M1 in seven rows of nine.

For reference, M1 evaluated at the harder full-vector target $\text{gap}/4$ used by the previous revision costs exactly four times these values, since the cost scales as the inverse square of the radius.

9.1 What sets the M1 cost

M1 must reach its common radius for every generator simultaneously, so its cost is set by the pool member with the largest reconstruction variance $\sum_{\alpha} \sigma_{i\alpha}^2$. In five of the nine cases that generator has a gradient that is zero to numerical precision, and its variance dwarfs the winner’s: for LiH it is $\mathbf{D} \ 0,1 \rightarrow 4,9$ with $\sum_{\alpha} \sigma_{i\alpha}^2 = 11.76$ against 0.072 for the actual winner, a factor of 164. In the remaining cases the binding generator carries a small but non-zero gradient, and on H₄ side=1.0 with the CISD state it is the winner itself.

This is not a defect of the comparison. M1 is non-adaptive, so it cannot know in advance which generators are irrelevant and must bound every one of them tightly enough to rule it out. Paying for high-variance operators that turn out not to matter is precisely the inefficiency that candidate elimination removes, and it is the mechanism behind the M1/M3 column.

It is also not the whole story. Restricting M1 to the generators with non-zero gradients — information no non-adaptive method has access to — reduces its cost by a factor of only 1.6 to 2.4 across the nine cases, because the remaining generators also carry substantial variance. The M1 cost is broadly distributed over the pool rather than concentrated in one irrelevant operator; the binding generator sets the constraint, but removing it would not make M1 competitive.

9.2 Where the cost actually sits: breadth and depth

The shot cost of a BAI method is not concentrated in the final elimination, as a naive reading of the $1/r^2$ scaling suggests. It splits into two terms of comparable size, and which one dominates changes with the system.

Breadth is the cost of sweeping the whole pool at the coarse radius set by the largest absolute-gradient deficit, so that the obvious losers can be dropped at all. *Depth* is the cost of the final tightening that separates the leader from its nearest rival. For M3 the split is visible directly in the per-context

accounting: a context that only ever served generators which were later eliminated keeps the shots it was given at the coarse radius, and those shots are still counted.

Table 5: Where each method’s shots are committed. For M2, the share paid by the two leading arms; for M3, the share of the final total already fixed at the first and coarsest breakpoint, when every generator is still active.

Case	M2: share paid by the top two arms	M3: share fixed at the coarsest breakpoint
H4 square, side=1.0 A, HF	53.8%	2.1%
LiH, R=3.0 A, HF	8.3%	87.7%
H2O stretch, R(OH)=1.75 A, HF	13.4%	60.7%

H₄ is depth-dominated: the last two breakpoints carry 94% of the M3 total. LiH and H₂O are breadth-dominated, and on LiH 88% of the M3 cost is already committed before a single generator has been eliminated.

The two terms reward sharing very differently. In the breadth term a shared context serves the entire pool, 92 or 140 generators at once, and M3’s advantage over M2 is correspondingly large. In the depth term it serves only the generators that survive to the final radius.

9.3 Why the M3 advantage decays with system size

Two quantities decide the depth term. The first is the fragmentation penalty

$$f_i = \left(\frac{\sum_{\alpha} \sigma_{i\alpha} \text{ over parent contexts}}{\sum_{\alpha} \sigma_{i\alpha} \text{ over its own contexts}} \right)^2, \quad (19)$$

the factor by which the shared grouping makes one gradient more expensive to resolve, which follows from Eq. (16) depending on $(\sum_{\alpha} \sigma_{i\alpha})^2$ rather than $\sum_{\alpha} \sigma_{i\alpha}^2$. The second is the number of generators still active at the final radius, written $|\mathcal{A}|$.

Table 6: Fragmentation of the winning gradient against the number of finalists. K is the number of fully commuting contexts the winner’s support occupies.

Case	State	K own	K parent	f	$ \mathcal{A} $	M2/M3
H4 square, side=1.0 A	HF	10	22	1.13	2	1.51
H4 square, side=1.0 A	CISD	7	28	1.07	3	1.91
H4 square, side=2.0 A	HF	10	22	1.19	3	1.92
H4 square, side=2.0 A	CISD	7	28	1.22	2	2.60
LiH, R=3.0 A	HF	58	520	3.49	3	1.79
H2O eq., R(OH)=0.9572 A	HF	57	858	2.54	2	1.13
H2O eq., R(OH)=0.9572 A	CISD	57	887	2.44	2	1.03
H2O stretch, R(OH)=1.75 A	HF	50	875	3.62	2	0.86
H2O stretch, R(OH)=1.75 A	CISD	50	899	2.94	2	1.01

The scaling is asymmetric. $|\mathcal{A}|$ is set by how many generators are nearly tied at the top, which has nothing to do with pool size and is two or three in every case here. The fragmentation penalty grows with the system, because a larger universal support spreads each individual gradient over more parent contexts: 22 on H₄, 520 on LiH, roughly 875 on H₂O. M3’s benefit in the depth term therefore has a ceiling while its cost in that term does not, and a crossover is unavoidable. Note that f grows far more slowly than the raw context count, 3.6 against 17.5 on H₂O stretch, because most parent contexts hold a negligible sliver of any one gradient.

9.4 The finalists do not share Pauli terms

The premise of universal grouping is that gradient operators share Pauli products. Across the pool they do: each Pauli product in $\mathcal{B}_0$ appears in roughly four different gradient commutators.

Table 7: Support sharing. Average reuse is $\sum_i |\text{supp}(C_i)|/|\mathcal{B}_0|$. Finalist overlap is the Jaccard overlap between the supports of the two leading generators.

Case	$ \mathcal{B}_0 $	$\sum_i \text{supp}(C_i) $	Average reuse	Finalist overlap
H4 square, side=1.0 A, HF	1,276	4,148	3.25	0.0%
LiH, R=3.0 A, HF	30,196	120,384	3.99	35.1%
H2O eq., R(OH)=0.9572 A, HF	81,566	314,720	3.86	0.0%
H2O stretch, R(OH)=1.75 A, HF	81,566	314,720	3.86	0.0%

The two columns tell opposite stories, and together they explain the whole pattern. Pool-wide reuse of about four is why M3’s breadth term is efficient. But in three of the four cases the two finalists share *no* Pauli products at all, so at the depth stage there is nothing to share and M3 pays the fragmentation penalty for nothing. LiH, the one case with substantial finalist overlap at 35%, is also the case where M3 retains a clear advantage despite the largest fragmentation penalty in the set.

9.5 An alternative: rebuild the grouping once the field narrows

If the fixed parent contexts are the problem at the depth stage, an obvious remedy is to abandon them once the active set is small and build a fresh fully commuting grouping over only the survivors’ union support, accepting that the earlier shots were taken in different circuit settings and do not transfer. This was tested directly. Phase one uses the parent contexts while more than a chosen number of generators remain active; phase two regroups from scratch and is charged the full cost of reaching the final radii, ignoring the head start phase one has already provided, so the totals below are upper bounds.

Table 8: Staged regrouping against the better of M2 and M3. The switch point is the active-set size at which the parent contexts are abandoned.

Case	Better of M2, M3	Best staged total	Switch at	Ratio
H4 square, side=1.0 A, HF	566,951	502,858	$ \mathcal{A} \leq 10$	0.89
LiH, R=3.0 A, HF	638,423	638,423	never	1.00
H2O stretch, R(OH)=1.75 A, HF	17,423,631	16,880,737	$ \mathcal{A} \leq 3$	0.97

Staged regrouping does beat both methods, but only by 3 to 11 per cent, and the best switch point is not consistent: early on H₄, late on H₂O, and never on LiH, where the procedure degenerates to M3. The reason is visible in Table 5. Regrouping can only improve the depth term, and on the systems where M3 struggles the cost is dominated by breadth, which the remedy does not touch. On H₂O stretch it improves M3 by 16 per cent and the better of the two methods by 3 per cent.

One incidental observation is more interesting than the headline. On H₄ the best switch point is early, when twenty generators still remain, and the regrouping of their union support needs only twelve contexts against the parent grouping’s fifty-three. Eliminating a handful of generators collapses the union support disproportionately, because the zero-gradient generators contribute a large share of $\mathcal{B}_0$. If regrouping is pursued, that is where the leverage is: not at the final pair, but as soon as the first substantial group of candidates has been ruled out.

10 Attribution: the no-sharing baseline

The three methods are two binary choices, sharing and elimination, and they span only three of the four combinations. The fourth is a baseline that does neither: every generator is measured in its own fully commuting grouping, and every one of them is resolved to the same common radius used by M1. It is reported here as a diagnostic outside the method hierarchy, and it serves two purposes. It makes the two mechanisms separately attributable, and it is a sanity bound, since measuring every generator separately is always available as a fallback.

Table 9: Cost of the no-sharing baseline and the improvement each mechanism buys against it. Sharing only is baseline/M1, elimination only is baseline/M2, both is baseline/M3. The product column is what “both” would be if the two mechanisms acted independently, and the interaction is the ratio of the two.

Case	State	Baseline	Sharing only	Elimination only	Both	Product	Interaction
LiH, R=3.0 A	HF	6,819,980	1.91	5.98	10.68	11.44	0.93
H4 square, side=1.0 A	HF	4,848,846	2.16	5.67	8.55	12.25	0.70
H4 square, side=1.0 A	CISD	130,375	2.19	1.27	2.43	2.77	0.88
H4 square, side=2.0 A	HF	2,002,941	3.35	4.74	9.09	15.87	0.57
H4 square, side=2.0 A	CISD	180,604	3.11	2.08	5.40	6.47	0.84
H2O eq., R(OH)=0.9572 A	HF	204,022,319	1.19	8.13	9.22	9.63	0.96
H2O eq., R(OH)=0.9572 A	CISD	485,001,056,522	1.18	10.44	10.78	12.30	0.88
H2O stretch, R(OH)=1.75 A	HF	4,407,619,149	1.14	252.97	218.59	288.50	0.76
H2O stretch, R(OH)=1.75 A	CISD	6,317,143,181	1.11	37.53	38.09	41.49	0.92

Two things follow, and both bear on how the project should be described.

The mechanisms are substitutes, not complements. The interaction is below unity in every one of the nine cases, between 0.57 and 0.96. Applying both mechanisms buys less than the product of applying each alone. This is the quantitative form of the argument in Sec. 9.2: elimination removes precisely the situation that sharing exploits, namely many generators active at once, so the second mechanism applied always finds less left to gain.

Sharing contributes little, and less as the system grows. On its own, universal grouping is worth a factor of 2.2 to 3.4 on H₄, 1.9 on LiH, and only 1.1 to 1.2 on H₂O. Elimination on its own is worth between 5 and 253. The headline mechanism of this project is the change of statistical target, not the shared measurement contexts; the latter matters mainly through the breadth term, and its contribution is fading with system size.

11 Finite-shot behaviour

Every number above is an oracle planning bound. The elimination rule reads exact gradients, so it cannot misfire, and the family-wise confidence is asserted rather than measured. This section replaces the exact gradients with estimates built from a finite number of shots and runs the same elimination rule on those.

Estimator fluctuations are drawn in the normal limit using the exact fragment variances, and shots accumulate across rounds rather than being redrawn, so successive rounds stay correlated as real data would be. Two approximations are deliberate. The normal limit is assumed rather than tested. And fluctuations are drawn independently per generator, which is exact for the baseline and for M2, where every generator has its own shots, but neglects the correlation between generators read off the same shots in a shared context for M1 and M3. That correlation affects how errors on different generators co-occur, not the per-generator rate, because the elimination rule uses individual radii.

Table 10: Realised cost against the planning bound. 200 trials for LiH and H₄, 40 for H₂O. The non-adaptive baseline and M1 realise their bound exactly, having no schedule to overshoot, and are omitted.

Case	State	M2: BAI-FC-IG			M3: BAI-FC-UG			M2/M3	
		Bound	Realised	Infl.	Bound	Realised	Infl.	Bound	Realised
LiH, R=3.0 A	HF	1,140,583	3,599,626	3.16	638,423	882,018	1.38	1.79	4.08
H4 square, side=1.0 A	HF	854,629	1,927,691	2.26	566,951	1,256,907	2.22	1.51	1.53
H4 square, side=1.0 A	CISD	102,806	307,780	2.99	53,727	151,805	2.83	1.91	2.03
H4 square, side=2.0 A	HF	422,746	968,209	2.29	220,389	552,263	2.51	1.92	1.75
H4 square, side=2.0 A	CISD	86,857	240,467	2.77	33,432	98,778	2.95	2.60	2.43
H2O eq., R(OH)=0.9572 A	HF	25,096,606	78,909,153	3.14	22,130,852	34,283,291	1.55	1.13	2.30
H2O stretch, R(OH)=1.75 A	HF	17,423,631	48,113,816	2.76	20,163,542	34,200,358	1.70	0.86	1.41

11.1 The one inversion does not survive

On H_2O stretch with the Hartree–Fock reference, the planning bounds give $\text{M2}/\text{M3} = 0.86$, the only row in which M3 is more expensive than M2. Under finite shots the same row gives 1.41, and M3 wins. The inversion was an artefact of comparing two bounds that inflate by different amounts, and the conclusion drawn from it in the previous revision is retracted in Sec. 16.

11.2 The two BAI methods inflate differently

The inflation is not a common factor. On H_4 , where the cost is depth-dominated, both methods inflate by 2.2 to 3.0 and the ratio between them is preserved to within a few per cent. On LiH and H_2O , where the cost is breadth-dominated, M2 inflates by 2.8 to 3.2 while M3 inflates by only 1.4 to 1.7, and the $\text{M2}/\text{M3}$ ratio roughly doubles.

The reason follows directly from Sec. 9.2. A real run does not know the deficits Δ_i in advance, so it discovers them by shrinking a common radius and overshooting each elimination point. M3’s breadth term is a single coarse sweep of the whole pool, a fixed cost that a schedule cannot overshoot, whereas every one of M2’s per-generator costs is discovered by overshooting. The more breadth-dominated the system, the less of M3’s cost is exposed to the schedule. Universal contexts therefore help more under finite shots than the planning bounds suggest, and the advantage grows with system size rather than shrinking.

11.3 The confidence model over-delivers by four orders of magnitude

No trial in any run selected the wrong generator. Sweeping the family-wise failure probability over nearly its whole range barely changes that.

Table 11: Calibration sweep on H_4 side=1.0, HF, 300 trials per entry. The stated δ has almost no influence on the realised selection-error rate.

δ	z_δ	M2 correct	M2 shots	M3 correct	M3 shots
0.05	3.10	100.0%	1,891,179	100.0%	1,324,843
0.40	2.42	100.0%	1,232,695	100.0%	849,844
0.90	2.11	99.3%	1,128,410	99.7%	691,975
0.99	2.07	100.0%	969,099	99.7%	741,804

The reason is structural rather than a defect of the simulation. The procedure stops when the common radius reaches $\text{gap}/2$, at which point the standard error of each estimate is $\text{gap}/(2z_\delta)$. The leader’s margin over the runner-up is one gap, so in units of the standard deviation of their difference it is $2z_\delta/\sqrt{2} = z_\delta\sqrt{2}$, and the selection-error probability is approximately

$$P(\text{wrong selection}) \approx 2\bar{\Phi}(z_\delta\sqrt{2}), \quad (20)$$

which is 1.6×10^{-1} at $z_\delta = 1$, 4.7×10^{-3} at $z_\delta = 2$ and 1.2×10^{-5} at the $z_\delta = 3.1$ used in the tables. The Bonferroni term enters only through z_δ , and the stopping radius already supplies most of the protection. The observed rates match Eq. (20) rather than δ .

The practical consequence is that δ as used does not control the selection-error rate. Inverting Eq. (20) for a genuine five per cent error suggests $z_\delta = 1.386$ against the 3.10 used in the tables, a factor of five in the planning bound. Simulation does not bear that out.

Table 12: Realised error and cost against the confidence factor, H_4 side=1.0, HF, 500 trials per entry.

z_δ	Eq. (20)	M2 error	M2 shots	M3 error	M3 shots
1.39	0.049	12.2%	1,586,040	10.6%	989,854
1.80	0.011	1.8%	1,113,341	2.0%	819,684
2.10	0.003	0.4%	1,211,688	0.0%	685,567
3.10	0.000	0.0%	2,004,737	0.0%	1,335,779

Eq. (20) accounts only for the final comparison and ignores generators wrongly eliminated earlier in the run, so it underestimates the error at small z_δ : it predicts five per cent at $z_\delta = 1.39$ where the simulation gives ten to twelve. The cost is also not monotone. Below $z_\delta \approx 1.8$ it rises again, because bad eliminations prolong the run. There is an optimum near $z_\delta \approx 1.8$ to 2.1, worth about 1.7 in realised cost at an error rate of two per cent or less — a real saving, but not the factor of five the bound suggested. Both conventions are selectable in the implementation; the tables in this report retain the Bonferroni one.

11.4 Good-enough selection

Stopping as soon as every surviving candidate can be certified within a tolerance Δ of the leader, rather than resolving the leader exactly, costs less for no practical loss. On H_4 side=1.0 HF at $\Delta = \text{gap}/4$, M2 falls from 2,007,147 to 1,359,837 shots and M3 from 1,282,967 to 893,811, both about thirty per cent, and in every trial the generator returned was still the exact maximiser.

12 Parameter sensitivity

12.1 What removing the two free parameters was worth

The previous revision evaluated M1 at a target radius of $\text{gap}/4$ with no task-level justification, and evaluated the BAI methods on a geometric schedule with shrink factor $\gamma = 0.5$. Both have been replaced. The M1 target is now $\text{gap}/2$, the loosest common radius at which a non-adaptive estimate separates the winner from the field, so that all three methods answer the same question; this divides the M1 column by four. The BAI methods are evaluated in the schedule-free limit; a coarse geometric schedule overshoots each elimination point and overpays.

The schedule change is not cosmetic. Under $\gamma = 0.5$, M3 cost more than M1 on three H_4 rows, violating the ordering the method guarantees by construction. That violation was entirely a discretisation artefact and disappears in the exact limit.

Table 13: Convergence of a geometric radius schedule onto the exact limit, H_4 side=1.0, HF.

γ	0.5	0.7	0.85	0.97	exact limit
M2: BAI-FC-IG	1,494,274	1,267,755	1,080,113	885,115	854,629
M3: BAI-FC-UG	928,366	872,427	712,219	584,142	566,951

For M2 the limit is the closed form

$$M_{M2} = \sum_i \left(\frac{z_\delta \sum_\alpha \sigma_{i\alpha}}{\Delta_i/2} \right)^2, \quad (21)$$

with Δ_i replaced by the gap for the leader. For M3 the trajectory has one breakpoint per distinct Δ_i and the allocation of Eq. (17) is evaluated at each.

12.2 The M1 target radius, if it is varied anyway

M1 has no internal criterion for its own precision, so it is worth recording how the comparison would move if the target were chosen differently. M2 and M3 are unaffected, their targets being fixed by the deficits.

Table 14: M1: FC-UG cost and the M1/M2 ratio against a target radius of gap/d . $d = 2$ is the identification criterion used in Table 4; $d = 4$ was the previous revision’s choice.

d	1	2	4	8	16
H4 side=1.0, HF: M1	561,216	2,244,862	8,979,447	35,917,787	143,671,148
M1/M2	0.66	2.63	10.51	42.03	168.10
H4 side=1.0, CISD: M1	14,911	59,642	238,565	954,260	3,817,039
M1/M2	0.15	0.58	2.32	9.28	37.13

12.3 The confidence level does not change the comparisons

All three methods scale with z_δ^2 , so every ratio in Table 4 is invariant under the family-wise failure probability. Verified for $\delta \in \{0.2, 0.05, 0.01, 0.001\}$: the H_4 side=1.0 HF ratios stay at 2.63 and 1.51 to better than one part in 10^3 , while the absolute costs move by about a factor of two. What δ does to the *realised* error rate is a separate matter, treated in Sec. 11.

13 Routine independence

Every number above comes from one first-fit grouping with one insertion order. First-fit is deterministic given an order, but the order is a free convention, so the conclusions have to be tested against it. The insertion order is now a parameter: three deterministic conventions (by decreasing weight, by increasing weight, and lexicographic) and a family drawn from a seed.

Table 15: Spread of the method ratios across groupings of the same problem. H_4 uses seven groupings and LiH four, each recomputing the baseline and all three methods. The H_2O row varies the parent grouping only, over four orderings.

Case	Contexts	M1/M2	M2/M3	M1/M3	Baseline/M3
H4 square, side=1.0 A, HF	49–61	1.91–2.63	1.51–1.97	3.35–4.55	8.47–12.27
LiH, R=3.0 A, HF	980–1,127	2.58–3.12	1.79–2.11	5.22–5.58	10.68–12.68
H2O stretch, R(OH)=1.75 A, HF	2,356–2,870	—	0.74–1.30	174–262	—

On H_4 and LiH the ratios move by at most a factor of 1.45 and the ordering of the methods is identical under every grouping tested. Those conclusions are routine-independent.

H_2O is not. The M3 cost varies by a factor of 1.77 across parent groupings, from 1.34×10^7 to 2.37×10^7 , and the M2/M3 verdict changes sign: M3 is cheaper under two of the four orderings and more expensive under the other two. **The M2 versus M3 comparison on H_2O is therefore not resolved.** It lies inside the spread induced by an arbitrary convention, and no verdict should be claimed in either direction — neither the planning-bound result that M2 wins there, nor the finite-shot result of Sec. 11 that M3 does. The M1 versus M3 comparison on the same rows is never in doubt, at 174 to 262.

14 Within-context correlation

The finite-shot study draws fluctuations independently per generator. That is exact for the baseline and M2, which give every generator its own shots, but M1 and M3 read several generators off the same

shots in a shared context. The covariance of two generators’ fragments inside one context follows from variances alone,

$$\text{Cov}(F_{i\alpha}, F_{j\alpha}) = \frac{1}{2} [\text{Var}(F_{i\alpha} + F_{j\alpha}) - \text{Var}(F_{i\alpha}) - \text{Var}(F_{j\alpha})], \quad (22)$$

and sampling from those matrices makes the simulation exact for the shared methods as well. It also permits a covariance-aware elimination rule that compares each pair at the confidence radius of their difference rather than at their individual radii.

Table 16: M3 under correct joint noise, 250 trials. “Marginal” is the elimination rule of this report; “pairwise” compares each pair at the radius of its difference.

Case	Noise model, rule	Shots	Correct	Against independent
H4 square, side=1.0 A, HF	independent, marginal	1,237,705	100.0%	—
	correlated, marginal	1,250,292	100.0%	1.01
	correlated, pairwise	624,963	100.0%	0.51
H4 square, side=1.0 A, CISD	independent, marginal	150,454	100.0%	—
	correlated, marginal	146,837	100.0%	0.98
	correlated, pairwise	92,431	100.0%	0.61
H4 square, side=2.0 A, HF	independent, marginal	551,541	100.0%	—
	correlated, marginal	3,369,764	79.2%	6.11
	correlated, pairwise	2,328,707	46.8%	4.22

Two readings, one of them incomplete.

Where the correlation is benign, modelling it changes the cost by one or two per cent, and the covariance-aware rule then cuts the realised cost of M3 by a further factor of two at no loss of accuracy. That is a clean gain, and it is the mechanism Part II proposes, arriving here as a saving rather than an optimisation.

H₄ side=2.0 with the Hartree–Fock reference behaves differently and is not yet understood. Correct selection falls to 79% under the marginal rule and to 47% under the pairwise one, with costs rising several-fold. The obvious explanation, that the leading pair is anticorrelated so that resolving each gradient fails to resolve their difference, is contradicted by direct measurement: the ratio $\text{Var}(\hat{g}_1 - \hat{g}_2)/[\text{Var}(\hat{g}_1) + \text{Var}(\hat{g}_2)]$ is 0.82 on that case and below one on all three, so the difference is easier to resolve than independence would imply, not harder. Correcting the confidence factor over pairs rather than over generators does not recover the accuracy either. Until this is explained, the correlated noise model is not used for reported numbers, and the independent figures stand with their stated approximation.

15 Relationship to the previous revision

The physics of the previous revision is confirmed. Universal Pauli supports, gradient values, gaps and selected generators are reproduced to six or more digits by an independent implementation. What changed is the cost model, for three reasons.

First, the fully commuting grouping. All rows now use one deterministic greedy routine; the previous revision’s H₄ and H₂O rows used a different partitioning, giving 356 and 9,874 contexts where greedy gives 53 and 2,366. Cost grows with the number of fragments an operator is split across, so this alone is a factor of several.

Second, the previous M1 column was not a single model. For LiH it is reproduced within 8% by a uniform per-context allocation at one standard error sized to a *typical* rather than worst-case generator (1,929,181 against the reported 2,088,380), which guarantees the target precision for about half the gradient vector rather than all of it. For H₄ no combination of grouping, allocation and confidence convention reproduces the reported 182,282,619: uniform allocation at worst-case variance with the previous revision’s own 356 contexts lands near 3×10^7 . That entry also exceeds its own no-sharing baseline of 11,326,894 by a factor of 16, which no valid allocation over shared contexts can do.

Third, the allocation itself. Solving Eq. (17) rather than spreading shots uniformly is worth about a factor of 17 on LiH; at one standard error the present M1 is *cheaper* than the previously reported value, 1,191,917 against 2,088,380, despite covering the worst case rather than a typical generator. The larger absolute values in Table 4 come principally from applying a Bonferroni family-wise z over the pool, a conservative convention applied identically to all three methods and therefore invisible in the ratios.

The consequence is that the previous M1/M2 ratios, which ranged from 0.28 to 627, should not be used. They combine an optimistic LiH entry with an unexplained H₄ entry.

16 Conclusions from the current data

1. M1, M2, and M3 are defined consistently and implemented as separate modules that save their FC groups, fragment variances and, for the BAI methods, elimination histories. All three are evaluated at one criterion, winner identification at family-wise $\delta = 0.05$, with no free schedule or precision parameter in any column.
2. **M3 is cheaper than M1 in every case**, by factors of 1.1 to 192. The ordering is guaranteed by construction; the content is the size of the factor, which grows with system size.
3. **M3 is cheaper than M2 in every case except one, and that one is unresolved.** It wins in eight rows of nine on the planning bounds. The exception, H₂O stretch with the Hartree–Fock reference, gives $M2/M3 = 0.86$ on the bounds and 1.41 under finite shots, but Sec. 13 shows the same quantity ranges over 0.74 to 1.30 across grouping routines. *No verdict should be claimed on that row in either direction*, and the previous revision’s conclusion that M3 is not always cheaper than M2 rested on a single arbitrary convention.
4. **M2 is not always cheaper than M1.** It wins in seven rows of nine; the counterexamples are both H₄ CISD rows, where the gap is wide and static estimation is already cheap.
5. **The two mechanisms are substitutes, not complements.** Against the no-sharing baseline the interaction term is below unity in all nine cases, between 0.57 and 0.96 (Sec. 10). Elimination removes the situation that sharing exploits, so applying both buys less than the product of applying each alone.
6. **Elimination is the dominant mechanism; sharing contributes little.** On its own, universal grouping is worth 2.2 to 3.4 on H₄, 1.9 on LiH, and only 1.1 to 1.2 on H₂O, while elimination alone is worth between 5 and 253. The change of statistical target, not the shared measurement contexts, is what carries the result.
7. **Finite shots favour the shared contexts.** M2 and M3 inflate above their planning bounds by different amounts, and the difference tracks breadth-dominance: on H₄ both inflate by 2.2–3.0 and the ratio is preserved, while on LiH and H₂O, M2 inflates by 2.8–3.2 against M3’s 1.4–1.7. M3’s breadth term is a fixed coarse sweep that a radius schedule cannot overshoot; every one of M2’s per-generator costs is discovered by overshooting. Sharing therefore helps more in practice than the bounds suggest, and increasingly so with system size.
8. **The confidence convention does not control the selection-error rate, and the obvious correction overshoots.** No trial in any run selected the wrong generator, and sweeping δ from 0.05 to 0.99 moves the realised rate only to 0.3–0.7%. But inverting Eq. (20) for a five per cent target gives ten to twelve per cent in simulation, because that expression ignores generators eliminated wrongly earlier in the run. The cost is not monotone in z_δ either. An optimum sits near $z_\delta \approx 1.8$ –2.1, worth about 1.7 in realised cost at two per cent error. Absolute costs should be read in that light; the comparisons between methods are unaffected.

9. **Conclusions 1 to 7 are routine-independent on H₄ and LiH** (Sec. 13), where ratios move by at most 1.45 across seven and four groupings respectively and no ordering changes.
10. **Modelling within-context correlation is a saving where it is understood, and an open question where it is not** (Sec. 14). On two of three H₄ cases it changes cost by one or two per cent and the covariance-aware elimination rule then halves M3’s realised cost at no loss of accuracy. On the third it degrades accuracy sharply for reasons not yet established, so the independent noise model of Sec. 11 remains the reported one.
11. The physics of the previous revision is confirmed; its cost model is not reproducible and its M1/M2 ratios should not be used. See Sec. 15.
12. All numbers remain single-ADAPT-step diagnostics. Nothing here speaks to the cost of a complete ADAPT trajectory, which is the metric Part III names as primary.

17 Interface with Parts II and III

Part II improves M3 by using measured variances and covariances to optimize overlapping universal operators and coefficient splitting. Sec. 9.2 gives that work a sharper target, but Sec. 9.4 also raises a caution that should be settled before it is built.

The mechanism Part II proposes for the decisive stage is the pairwise-variance equation of that note: best-arm identification only requires the pairwise differences to be resolved, and

$$\text{Var}(\hat{g}_i - \hat{g}_j) = \sum_{\alpha} \frac{1}{m_{\alpha}} \text{Var}(F_{i\alpha} - F_{j\alpha}), \quad (23)$$

so the covariance that helps enters through co-measurement inside a shared context, not through the two generators sharing Pauli products. That premise was tested on the finalist pair of each system, inside the parent contexts, by comparing the cost of resolving the difference alone against the cost of resolving both generators to the same radius.

Table 17: Separating the two finalists inside the shared parent contexts: resolving only their difference against resolving each of them, both to radius gap/2. $\sum_{\alpha} \sigma$ is summed over the parent contexts each object touches.

Case	$\sum_{\alpha} \sigma_i$	$\sum_{\alpha} \sigma_j$	$\sum_{\alpha} \sigma_{i-j}$	Resolve both	Resolve difference
H4 square, side=1.0 Å, HF	1.921	1.731	2.455	554,857	499,910
LiH, R=3.0 Å, HF	2.170	2.180	3.623	48,712	67,577
H2O eq., R(OH)=0.9572 Å, HF	7.263	12.173	17.803	1,002,360	1,581,034
H2O stretch, R(OH)=1.75 Å, HF	6.239	7.794	13.005	10,636,302	18,049,318

Resolving the difference is cheaper only on H₄, by a factor of 1.11; on LiH and both H₂O rows it is more expensive, by factors up to 1.7. The arithmetic is unforgiving. Resolving two generators separately costs $\sigma_i^2 + \sigma_j^2$ in units of ϵ^{-2} , while resolving one difference observable costs σ_{i-j}^2 , so the difference wins only when $\sigma_{i-j} < \sqrt{\sigma_i^2 + \sigma_j^2}$, which for comparable arms demands roughly 30% cancellation. The measured cancellation, $1 - \sigma_{i-j}/(\sigma_i + \sigma_j)$, is 33% on H₄ but only 17% on LiH and 7% on H₂O stretch, consistent with the finalist support overlaps in Table 7.

This does not refute Part II, whose other mechanisms — coefficient splitting over an overlapping cover, empirically estimated context covariances, and adaptive choice of the next context to measure — are untouched by this test. It does mean the pairwise-difference mechanism should not be assumed to pay, and that whether cancellation is available must be measured on the intended benchmark before the

variance-aware machinery is designed around it. Sec. 9.2 adds a second caution: that mechanism acts on the depth term, which is only 12% of the M3 cost on LiH and about 39% on H₂O stretch, so its total effect is bounded even where it does pay.

Part III changes the ADAPT trajectory so that the commutators first become symmetry-filtered and tapered. When Part III is active, the Part I object C_i is replaced by

$$C_{i,\text{tap}}^{\parallel,k} = \text{Taper}_{\mathcal{R}_k}(\Pi_{\mathcal{R}_k}([H, G_i])), \quad (24)$$

and then M1, M2, or M3 is applied to these reduced operators. Part I tables do not include this reduction.

18 Implementation status

The following items of the previous revision’s cleanup list are complete.

1. Separate output files named with the method codes M1_FCUG, M2_BAIFCIG and M3_BAIFCUG.
2. FC groups, fragment standard deviations and BAI elimination histories saved for every number in every table.
3. Deterministic grouping conventions recorded with every run, together with the interpreter and package versions.
4. M3 completed for H₂O, so the headline comparison no longer has missing entries.

Two implementation defects of the previous revision are worth recording, because both were silent. The two-electron integrals were inserted into the spin-orbital tensor in chemist rather than physicist ordering, with the spin pattern (a, b, a, b) instead of (a, b, b, a) and without the one-half prefactor; and the computational basis was indexed little-endian where the sparse-operator convention is big-endian. For H₄ the resulting Hartree–Fock energy was -5.21 Ha against an RHF value of -1.76 Ha. The validation gate described in Sec. 6 exists to make a repeat of this impossible.

19 Remaining work in Part I

1. **Explain the H₄ side=2.0 correlation anomaly** (Sec. 14). Correct joint noise degrades selection accuracy on that one case, the leading explanation is refuted by measurement, and the covariance-aware rule does not recover it. Until this is understood the correlated noise model cannot be used for reported numbers, and the finite-shot figures keep an approximation that is exact only for the baseline and M2.
2. **Extend the routine sweep to the remaining H₂O rows.** One row has been swept and it dissolved a conclusion. The other three, and the individual groupings that M2 and the baseline depend on, have not.
3. **Adopt a confidence convention.** The evidence points to $z_\delta \approx 1.8$ to 2.1. Choosing it changes every absolute number and the tables should be regenerated once it is fixed.
4. **Make good-enough selection the default stopping rule** rather than a diagnostic, given that it costs thirty per cent less with no observed loss.
5. **Version control.** The raw commutator expansions are now saved with each run; the code still needs a repository so that a commit hash can accompany every table.

20 Reproduction

```
python -m pip install -r requirements.txt
python scripts/run_part1_methods.py --all --out runs
python scripts/run_finite_shot.py --case H4_square_eq_side1p0_HF \
    --trials 200 --out runs
python -m pytest tests -q
python scripts/verify_part1.py --runs runs
```

Each case writes a per-method summary together with the FC groups, fragment standard deviations and BAI histories behind it. The test suite pins the conventions that were previously wrong: the computational-basis ordering, the qubit Hamiltonian against PySCF RHF and FCI, the top gradient against a finite-difference energy derivative, and the context-shot allocation against its closed form for a single gradient.

`verify_part1.py` checks the published results rather than the implementation. For every case it confirms that the reported total is the sum of the file that claims to explain it, that the validation gate passed, that the allocation meets its confidence target, that M3 does not exceed M1, and that all methods agree on the selected generator; against the live code it confirms that the fragment decomposition sums back to the gradient, that repeated runs agree exactly, that the ratios are invariant to the confidence level, and that a geometric radius schedule converges onto the exact limit used in the tables. All 68 checks pass, alongside 26 unit tests.

References

- [1] H. R. Grimsley, S. E. Economou, E. Barnes, and N. J. Mayhall, “An adaptive variational algorithm for exact molecular simulations on a quantum computer,” *Nature Communications* **10**, 3007 (2019). <https://doi.org/10.1038/s41467-019-10988-2>
- [2] V. O. Shkolnikov, N. J. Mayhall, S. E. Economou, and E. Barnes, “Avoiding symmetry roadblocks and minimizing the measurement overhead of adaptive variational quantum eigensolvers,” *Quantum* **7**, 1040 (2023). <https://doi.org/10.22331/q-2023-06-12-1040>
- [3] P. G. Anastasiou, N. J. Mayhall, E. Barnes, and S. E. Economou, “How to really measure operator gradients in ADAPT-VQE,” *arXiv:2306.03227* (2023). <https://arxiv.org/abs/2306.03227>
- [4] R. Huang and A. F. Izmaylov, “Quantum Gambling: Best-Arm Strategies for Generator Selection in Adaptive Variational Algorithms,” *arXiv:2509.14917* (2025); *J. Chem. Theory Comput.* (2026). <https://arxiv.org/abs/2509.14917>
- [5] E. Rossi, E. R. Kjellgren, A. F. Izmaylov, S. P. A. Sauer, K. M. Ziems, and S. Coriani, “Resource-efficient energy-based operator selection in fermionic ADAPT-VQE via exact Hamiltonian transformation,” *arXiv:2606.04786* (2026). <https://arxiv.org/abs/2606.04786>